Tuning Panel Methods For Small Datasets

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Contents

1 Introduction

Motivation

Most of the phenomena we study can't be experimentally manipulated:

- How do minimum wage laws affect unemployment? (Card and Krueger 1993)
- How does media coverage affect voting patters? (DellaVigna and Kaplan 2007)
- How do political institutions affect Economic growth? (Acemoglu et al. 2004)

In these cases, we often use panel data methods.

- But these make strong assumptions!
  - Parallel Trends:

$$\mathbb{E}[Y_1(0) - Y_0(0) \mid D = 1] = \mathbb{E}[Y_1(0) - Y_0(0) \mid D = 0]$$

Motivation (ii)

Matrix Completion (Athey et al. 2021) and the Generalized Synthetic Control (Xu 2017) are becoming increasingly popular because they relax the assumptions needed for valid inference:

- Eibl and Hertog (2023) investigate the provision of public services in oil-rich countries
- Bischof and Wagner (2019) study how voters react to a radical party being seated in European parliaments
- Alrababa'h et al. (2021) estimate the effects of celebrity soccer players on hate crimes.
- Finnegan et al. (2025) examine why some governments are better at setting economic policies than others.

These methods are also increasingly popular because they give tight confidence intervals!

(Even in small datasets)

Running Example

Gilens et al. (2021) uses these methods to estimate the effect of the Citizens United Decision on corporate tax rates.

How did Citzens United affect tax rates in states that had bans on corporate spending bans?

- Yearly panel data 2000-2016
  - 10 years pre-treatment
  - 7 years post-treatment
- 32 states
  - 8 treated
  - 24 untreated (were not affected by Citizens United)
  - (18 states dropped from analysis)

Total of N=512 observations, 488 of which are untreated.

Running Example (ii)

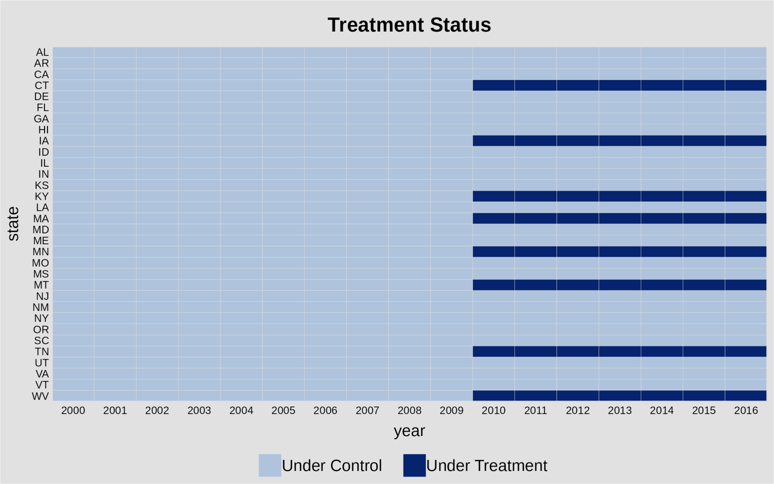


Figure 1: Treatment Uptake, Citizens United Decision

Running Example (iii)

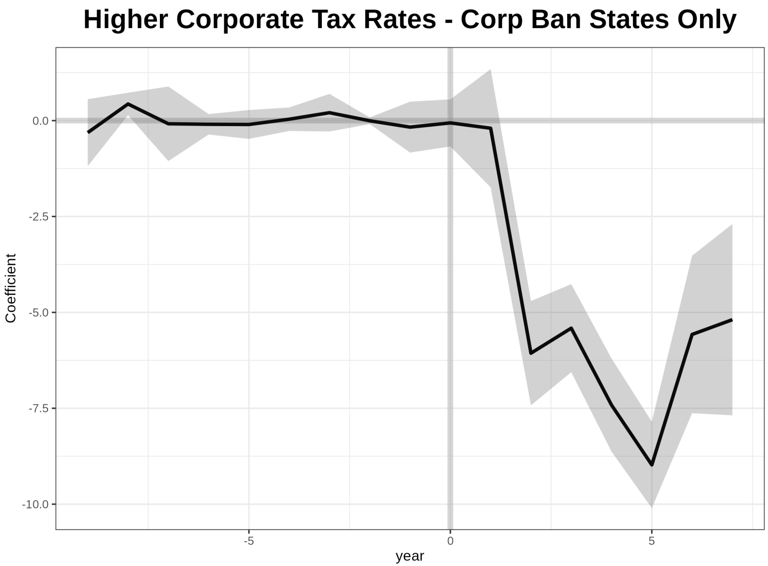


Figure 2: Effect of Citizens United on Corporate Tax Rate. Replicated from Gilens et al. (2021)

Running Example (iv)

Standard errors imply we could detect an effect of around a 1% change in corporate tax rates.

Remarkable because there were only 8 treated units

- Only three treated states changed tax rates post verdict
  - CT (increased taxes once)
  - MA (decreased taxes three times)
  - WV (decreased taxes four times)

Research Questions

Are researchers picking these estimates because they give misleadingly small confidence intervals?

Weiss (2024) points out that many other panel estimators are motivated by asymptotic arguments that may not apply in practice.

Question 1

Are inferences from these methods valid in small samples?

Question 2

Is there a way to ensure valid inference even outside of the asymptotic regime?

2 Theoretical Analysis

Theoretical

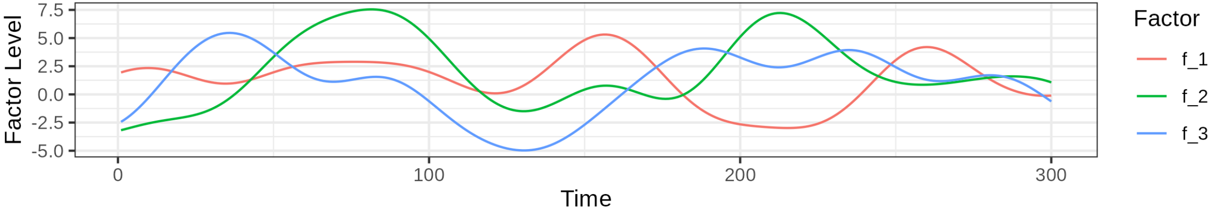
Gsynth / Matrix Completion estimates observed outcomes as:

$$Y_{\text{it}} = \underbrace{\delta_{\text{it}} D_{\text{it}}}_{\text{treatment effect}} + \underbrace{x_{\text{it}}^T \beta}_{\text{covariate adjsutment}} + \underbrace{\gamma_t + \eta_i}_{\text{two-way fixed effects}} + \underbrace{\lambda_i^T f_i}_{\text{latent factors}} + \underbrace{\varepsilon_{\text{it}}}_{\text{shocks}}$$

What is a latent factor?

Unobserved time trends that individual units ‘load’ on with ‘loadings’ λ_i

Examples:



Theoretical (ii)

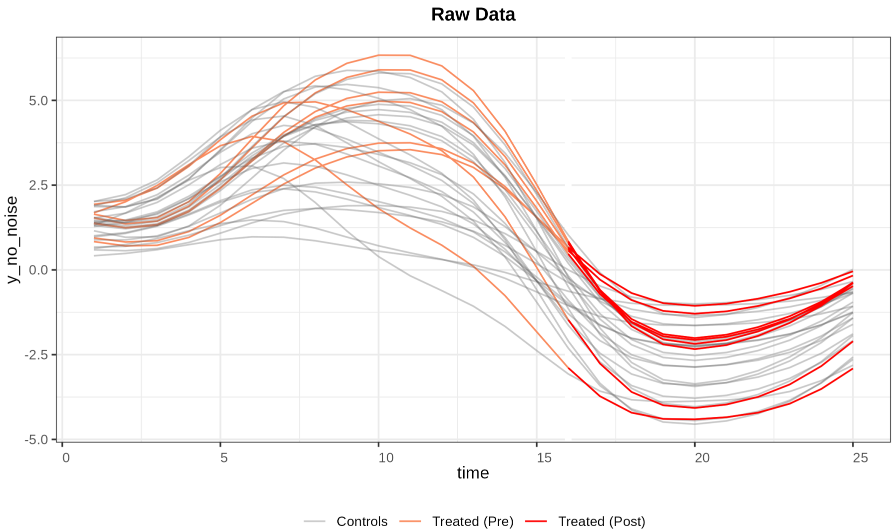


Figure 4: Latent Factors allow for units to evolve differentially over time

Theoretical (iii)

Fundamental difference from OLS: parameters are estimated with **bias** that decreases with sample size:

- Expect bias in small samples if the matrix FF^T is ill-conditioned

Estimate variance with bootstrap that estimates *conditional* variance:

$$\mathrm{Var}\left[\widehat{\mathrm{ATT}} \mid \hat{\beta}, \hat{\gamma}, \hat{\eta}, \hat{\lambda}, \hat{f}\right] \neq \mathrm{Var}\left[\widehat{\mathrm{ATT}}\right]$$

- Under-estimation of variance when uncertainty about $\beta, \gamma, \eta, \lambda, f$ is a large component about our uncertainty about $\widehat{\mathrm{ATT}}$
  - In run Gilens et al. (2021) analysis $|\beta| + |\gamma| + |\eta| + |\lambda| + |f| = 108$

3 Monte-Carlo Evidence

Monte-Carlo Evidence - Bias

Sample size:

- 15 time periods before and after treatment
- 40 control units
- 10 treated units

DGP details:

- 2 latent factors:
  - $f_1(t) = \log(t * 2)$
  - $f_2(t) = \sqrt{t}$
- Factor loadings drawn from Dirichlet$(2, 1)$ for treated units, and Dirichlet $(1, 2)$ for control units
- No Fixed effects, treatment effect, or covariates
- $\varepsilon_{it} \sim N\left(0, \sqrt{.4}\right)$

Monte-Carlo Evidence - Bias (ii)

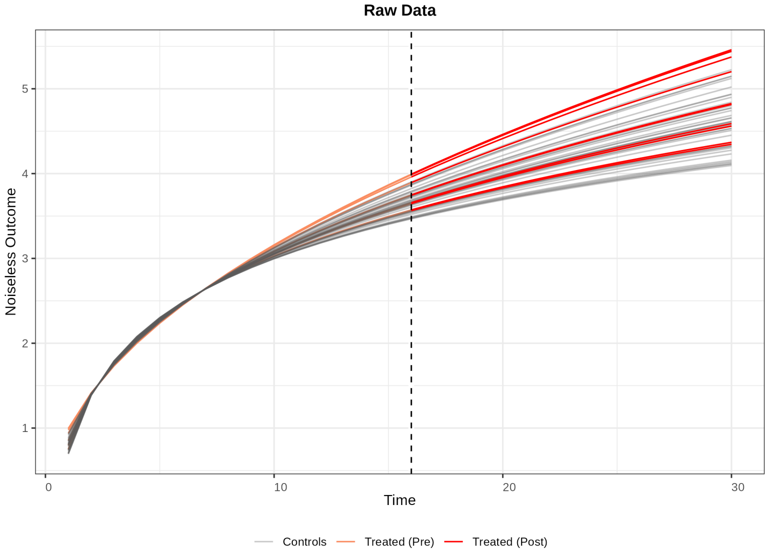


Figure 5: Simulated Dataset - No Noise

Monte-Carlo Evidence - Bias (iii)

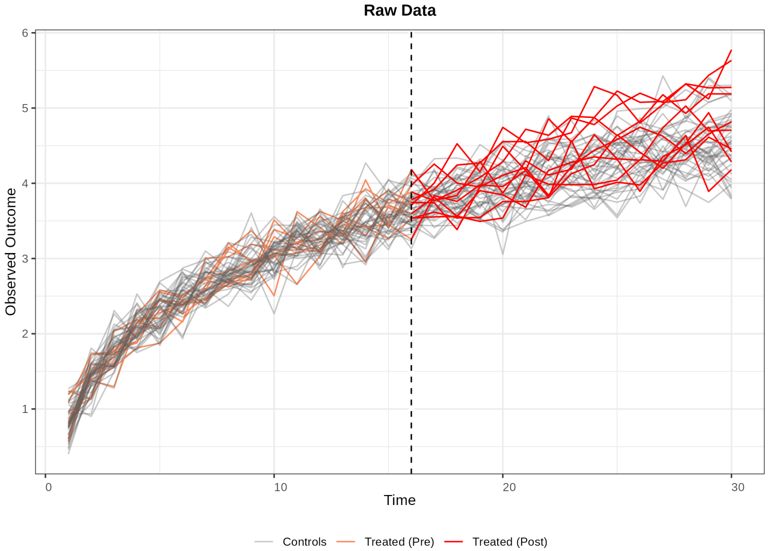


Figure 6: Simulated Dataset - Observed Outcomes

Monte-Carlo Evidence - Bias (iv)

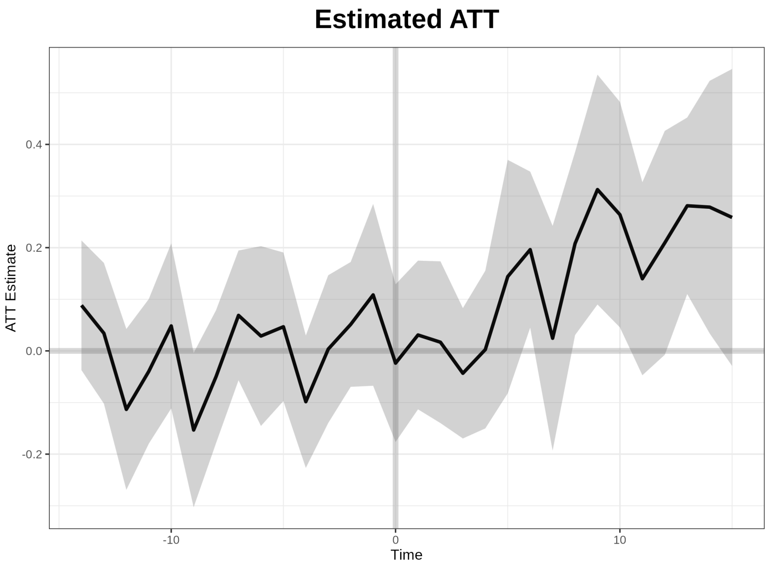


Figure 7: Simulated Dataset - Observed Outcomes

Monte-Carlo Evidence - Bias (v)

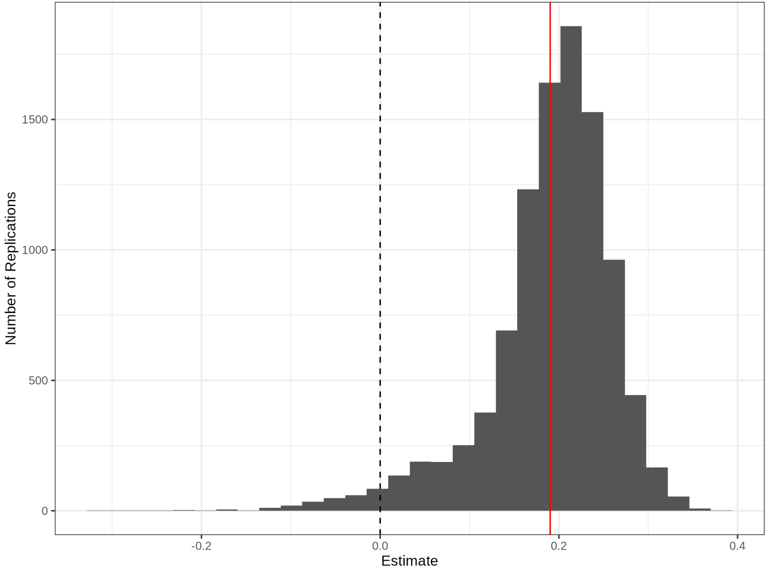


Figure 8: Bias of GSC Estimator

Monte-Carlo Evidence - Bias (vi)

- Potential for extreme bias in finite samples when FF^T is ill-conditioned
- Model assumptions are satisfied, but we have $\approx 15\%$ coverage
- At odds with results from Xu (2017), which showed good coverage and low bias
  - Latent factors very easy to identify in those settings, $\kappa(FF^T) \approx 1$

Bias is a problem, but what about the mismatch between conditional variance and variance?

Monte-Carlo Evidence - Variance

Replicate simulations from Xu (2017), but across different levels of factor overlap and sample sizes.

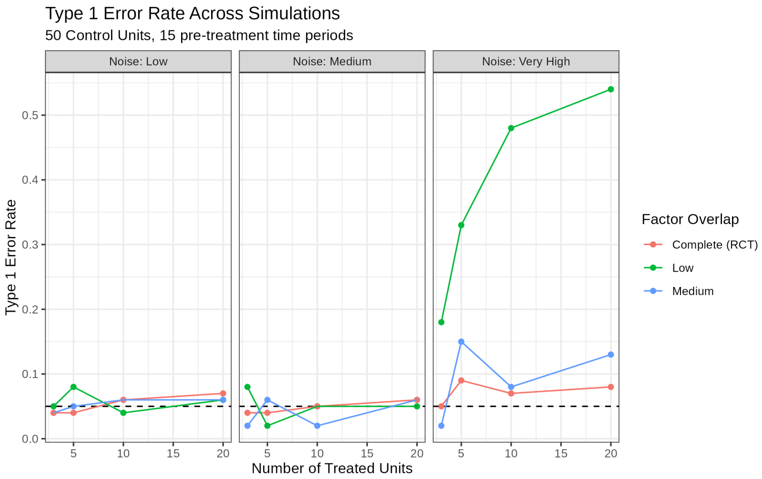


Figure 9: Coverage of GSC in Auxiliary simulations

4 Potential Solution

A Proposed Solution

So far, we have seen that:

1. The finite-sample bias of these estimators can be substantial
2. Coverage can be poor in small-samples

Is there a way to get valid inference even when N is small? Yes!

- Existing bootstrap approach gives consistent estimates of $\mathrm{Var}\left[\widehat{\mathrm{ATT}} \mid D, X, \Lambda, F\right]$
  - Assumes we know nuisance parameters
  - Ignores finite-sample bias

We will calibrate variance estimates using a double-bootstrap approach

Double Bootstrap Approach

Double Bootstrap (Aronow et al. 2025; Beran 1988)

Fit a distribution $\hat{F}$ to predict control outcomes
Initialize array a
for i in $\{0 \ldots N\}$
  Draw a sample from $\hat{F}$
  Fit GSC to sample, compute standard errors with bootstrap
  $a[i] \leftarrow \frac{\widehat{\mathrm{ATT}}}{\widehat{\mathrm{se}}\left(\widehat{\mathrm{ATT}}\right)}$
  end
Compute Variance Inflation Factor, $\hat{\beta} = \min_{\beta} : \sum_{i=1}^{N} I(|\beta \times a[i]| < 1.96) < 0.05 * N$
Estimate GSC on original data, inflate estimated variance by $\max\left(1, \hat{\beta}\right)$
end

Double Bootstrap Approach (ii)

- Why is this useful? Ensures we have exact coverage under at least one distribution, $\hat{F}$ (If you use sample-splitting for $\hat{F}$).
  - Inflates standard errors to correct for finite-sample bias, which is not handled for classic bootstrap
  - Also factors in uncertainty about nuisance parameters

We can use any estimate for $\hat{F}$, even an inconsistent one! The motivation is just to correct coverage under a plausible similar DGP.

- Block Bootstrap methods
- Flexible Nonparametric Bayesian models (proposed approach)

Simulate distribution over which estimator performs poorly, then correct it to perform well.

5 Future Directions

Moving Forward

So far:

- Conducted simulation studies to try and quantify bias + under-coverage issue
- Working on double bootstrap approach
  - Efficiency is a major issue because bootstrap is used twice

Next steps

- Extend to clustering
- Investigate the effects of serially-correlated errors, heteroskedasticity

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